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Dual ring stuffing box

Abstract

A system includes a fluid end having a fluid end face and a recess extending toward a base. The system also includes a pressure ring extending from the fluid end face, the pressure ring having a pressure ring bore diameter larger than a fluid end bore diameter. The system further includes an adjacent ring coupled to the fluid end, the pressure ring positioned between the adjacent ring and the base, wherein at least one fastener extends through an aperture of the adjacent ring to couple the adjacent ring to the fluid end. The system includes packing material positioned within at least a portion of the pressure ring bore, the packing material positioned such that axial movement toward the fluid end is blocked by the base.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. Non-Provisional application Ser. No. 17/864,873, filed Jul. 14, 2022, titled “DUAL RING STUFFING BOX,” which is a continuation-in-part of U.S. patent application Ser. No. 17/213,854, filed Mar. 26, 2021, titled “DUAL RING STUFFING BOX,” now U.S. Pat. No. 11,391,374, issued Jul. 19, 2022, which claims the benefit of U.S. Provisional Patent Application No. 63/137,501, filed Jan. 14, 2021, titled “DUAL RING STUFFING BOX,” and is a continuation-in-part and claims the benefit of U.S. patent application Ser. No. 29/769,314, filed Feb. 4, 2021, titled “FLUID END FOR A PUMPING SYSTEM,” now U.S. Patent No. D933104, issued Oct. 12, 2021, and is a continuation-in-part and claims the benefit of U.S. patent application Ser. No. 29/769,318, filed on Feb. 4, 2021, titled “FLUID END FOR A PUMPING SYSTEM,” now U.S. Patent No. D933105, issued Oct. 12, 2021, the disclosures of which are incorporated herein by reference in their entireties for all purposes.

TECHNICAL FIELD

(1) Embodiments of the subject matter disclosed herein generally relate to pump systems, and in

particular to sealing assemblies for pumping systems.

BACKGROUND

(2) Pumping systems may be used in a variety of applications, such as industrial applications where pumping systems are used to elevate a working fluid pressure. One such application is hydraulic fracturing systems, where high pressure pumps are used to increase a fluid pressure of a working fluid (e.g., fracturing fluid, slurry, etc.) for injection into an underground formation. The working fluid may include particulates, which are injected into fissures of the formation. When the fluid is removed from the formation, the particulates remain and “prop” open the fissures, facilitating flow of oil and gas. In many applications, reciprocating pumps are used where a fluid is introduced into a fluid end inlet passage and out through an outlet passage. A plunger reciprocates within a bore to add energy to the fluid. A stuffing box may be positioned around the plunger to block fluid leaks.

SUMMARY

(3) Applicant recognized the problems noted above herein and conceived and developed embodiments of systems and methods, according to the present disclosure, for stuffing boxes.

(4) In an embodiment, a system includes a pressure ring positioned to abut a fluid end face, the pressure ring having a pressure ring bore with a diameter larger than a fluid end bore diameter. The system also includes an adjacent ring coupled to a fluid end, the pressure ring positioned between the adjacent ring and the fluid end face. The system further includes a face seal positioned at an interface of the pressure ring and the fluid end face, the face seal containing pressurized fluid associated with the fluid end. The system also includes packing material positioned within at least a portion of the pressure ring bore.

(5) In an embodiment, a system includes a fluid end, the fluid end having a recessed area extending to a fluid end face. The system further includes a pressure ring positioned within the recessed area against the fluid face to compress the face seal, the pressure ring having a pressure ring bore positioned co-axially with a fluid end bore, and the pressure ring bore has a diameter greater than the fluid end bore. The system also includes an adjacent ring positioned against the pressure ring and coupled to the fluid end. The system includes a face seal arranged between the pressure ring and the fluid end. The system further includes packing material positioned within the pressure ring bore, at least a portion of the packing material being positioned external to the fluid end bore.

(6) In an embodiment, a method includes inserting a face seal into a face groove. The method also includes positioning a pressure ring, within a recessed portion, to compress the face seal. The method further includes positioning an adjacent ring against the pressure ring. The method also includes securing the adjacent ring to a fluid face, wherein the pressure ring is positioned between the adjacent ring and the fluid end. The method includes installing, within a pressure ring bore, packing material.

(7) In an embodiment, a system includes a fluid end having a fluid end face and a recess extending toward a base. The system also includes a pressure ring extending from the fluid end face, the pressure ring having a pressure ring bore diameter larger than a fluid end bore diameter. The system further includes an adjacent ring coupled to the fluid end, the pressure ring positioned between the adjacent ring and the base, wherein at least one fastener extends through an aperture of the adjacent ring to couple the adjacent ring to the fluid end. The system includes packing material positioned within at least a portion of the pressure ring bore, the packing material positioned such that axial movement toward the fluid end is blocked by the base.

(8) In an embodiment, a system includes a fluid end, the fluid end having a recessed area. The system also includes a pressure ring extending from the recessed area, the pressure ring having a pressure ring bore positioned co-axially with a fluid end bore, and the pressure ring bore having a diameter greater than the fluid end bore. The system further includes an adjacent ring positioned against the pressure ring and coupled to the fluid end. The system includes packing material positioned within the pressure ring bore, at least a portion of the packing material being positioned external to the fluid end bore. The system also includes an injection port extending through the

adjacent ring, the injection port arranged in a low pressure region, axially away from a seal between the pressure ring and the adjacent ring.

(9) In an embodiment, a method includes providing a fluid end including a recessed area extending into a fluid end face and a pressure ring extending axially out of the recessed area. The method also includes positioning an adjacent ring against the pressure ring. The method further includes securing the adjacent ring to the fluid end. The method includes installing, within a pressure ring bore, packing material.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The present technology will be better understood on reading the following detailed description of non-limiting embodiments thereof, and on examining the accompanying drawings, in which:
- (2) FIG. 1 is a schematic cross-sectional view of a prior art pump assembly;
- (3) FIG. 2 is a cross-sectional view of an embodiment of a stuffing box assembly, in accordance with embodiments of the present disclosure;
- (4) FIG. 3 is an isometric view of an embodiment of a stuffing box assembly, in accordance with embodiments of the present disclosure;
- (5) FIG. 4 is a cross-sectional view of an embodiment of a stuffing box assembly, in accordance with embodiments of the present disclosure;
- (6) FIG. 5 is an isometric view of an embodiment of a stuffing box assembly, in accordance with embodiments of the present disclosure;
- (7) FIG. 6 is a cross-sectional view of an embodiment of a stuffing box assembly, in accordance with embodiments of the present disclosure;
- (8) FIG. 7 is an isometric view of an embodiment of a fluid end, in accordance with embodiments of the present disclosure;
- (9) FIG. 8 is an isometric view of an embodiment of a fluid end, in accordance with embodiments of the present disclosure;
- (10) FIG. 9 is an isometric view of an embodiment of a fluid end with a stuffing box assembly, in accordance with embodiments of the present disclosure;
- (11) FIG. 10 is an isometric view of an embodiment of a fluid end with a stuffing box assembly, in accordance with embodiments of the present disclosure;
- (12) FIG. 11 is a cross-sectional view of an embodiment of a stuffing box assembly, in accordance with embodiments of the present disclosure;
- (13) FIG. 12 is a cross-sectional view of an embodiment of a stuffing box assembly, in accordance with embodiments of the present disclosure;
- (14) FIG. 13 is a sectional isometric view of an embodiment of a stuffing box assembly, in accordance with embodiments of the present disclosure; and
- (15) FIG. 14 is a perspective view of an embodiment of a stuffing box assembly, in accordance with embodiments of the present disclosure.

DETAILED DESCRIPTION

(16) The foregoing aspects, features, and advantages of the present disclosure will be further appreciated when considered with reference to the following description of embodiments and accompanying drawings. In describing the embodiments of the disclosure illustrated in the appended drawings, specific terminology will be used for the sake of clarity. However, the disclosure is not intended to be limited to the specific terms used, and it is to be understood that each specific term includes equivalents that operate in a similar manner to accomplish a similar purpose.

(17) When introducing elements of various embodiments of the present disclosure, the articles “a”,

“an”, “the”, and “said” are intended to mean that there are one or more of the elements. The terms “comprising”, “including”, and “having” are intended to be inclusive and mean that there may be additional elements other than the listed elements. Any examples of operating parameters and/or environmental conditions are not exclusive of other parameters/conditions of the disclosed embodiments. Additionally, it should be understood that references to “one embodiment”, “an embodiment”, “certain embodiments”, or “other embodiments” of the present disclosure are not intended to be interpreted as excluding the existence of additional embodiments that also incorporate the recited features. Furthermore, reference to terms such as “above”, “below”, “upper”, “lower”, “side”, “front”, “back”, or other terms regarding orientation or direction are made with reference to the illustrated embodiments and are not intended to be limiting or exclude other orientations or directions. Moreover, like numbers may be used for like components, but such disclosure is for convenience purposes and is not intended to limit the scope of the present disclosure.

(18) Traditional fluid ends feature an integrated stuffing box which requires a larger forging and more material to be removed. This is an expensive process. An alternative is a removable and replaceable stuffing box where the pressure vessel components are attached to the fluid end. This has been achieved previously by stuffing boxes which were threaded into the fluid end via large power threads such as ACME or Buttress profiles. These threaded stuffing boxes were heavy and difficult to install and then prone to failures of the threaded interfaces.

(19) Embodiments of the present disclosure address and overcome problems faced with traditional stuffing boxes utilized with pumping units, such as fluid ends for fracturing pumps, and may be further incorporated in various systems with reciprocating components. In at least one embodiment, systems and methods of the present disclosure include a set of rings flanged into place such that a pressure containing ring is made from a wear resistant stainless steel, or any other type of wear resistant material, and an adjacent ring is made from a less expensive alloy steel, or any type of compatible material, as well as similar wear resistant materials. The two rings are joined together and preloaded by a series of fasteners (e.g., studs, nuts, bolts, cap screws, etc.). In various embodiments, the pressure containing ring houses a face seal or the fluid cylinder may house the face seal. Furthermore, the seal could be partially set into both. In one or more embodiments, a sealing element is arranged between the two rings, which includes a sealing component, such as an o-ring or other seal, which may be retained in the pressure ring, the adjacent ring, or both. Various embodiments also include a cross drilled weep hole where the intersecting holes may or may not be drilled entirely through the pressure ring. For example, in some iterations they may be intersecting blind holes. The pressure ring extends past the last pressure ring of a packing set such that anything closer to the exterior of the fluid end (e.g., radially outward from a pressure chamber) is not exposed to pressurized fracking fluid.

(20) In one or more embodiments, systems and methods use studs and nuts that pass through dual rings of the stuffing box. It should be appreciated that these fasteners are provided by way of example only and are not intended to limit the scope of the present disclosure, as other fasteners, such as bolts and the like, may be used. Moreover, a threaded or otherwise locking interface between the rings may be utilized, such as a bayonet connection, J-hook, or the like. The fluid end may be machined such that an outside diameter of the pressure ring serves as a fine alignment feature during installation. The pressure ring may also be arranged such that it is also the only portion of the stuffing box exposed to the high pressure fracking fluid. Accordingly, in various embodiments, the pressure ring may be formed from a high grade stainless steel, among other wear resistant and/or specialized materials, while the adjacent ring would be made from alloy steel, among other types of materials, thereby reducing costs associated with implementation of the various embodiments disclosed herein. In at least one embodiment, the pressure ring may be integrally formed or be continuous with the materials associated with the fluid end.

(21) Embodiments of the present disclosure may incorporate a sealing element within the fluid end

body and/or within the pressure ring. This configuration may further include a single outer diameter. As a result, a seal designed to seal against the pressurized fluid may be retained by the pressure ring and/or the adjacent ring. Furthermore, embodiments may include a grease insertion point associated with the adjacent ring and also one or more threads to receive a packing nut. Moreover, a recessed shelf may be incorporated such that radial holes in the packing nut are clearly exposed above the heads of the fasteners (e.g., studs and nuts, bolts, cap screws, etc.). In certain embodiments, the packing set rests on the fluid end block surface. In others, the pressure containing ring features a shelf where a spacer ring may be installed and retained such that the packing assembly sits on the spacer ring.

(22) In one or more embodiments, a section of removed material across a backside of the fluid end, such that the stuffing box does not sit within any bore but sits on the back face of the fluid end, is incorporated. For example, a recess or groove may be formed within the fluid end to receive the pressure ring. In one or more embodiments, the recess or groove may extend substantially across the entire fluid end. In certain embodiments, the recess or groove may be particularly located at individual locations for individual stuffing boxes. Furthermore, the pressure ring may extend from the fluid end face such that the groove or recess extends radially into the body of the fluid end. As will be described below, in various embodiments, the recess or groove may be squared or slanted. Furthermore, a depth of the recess or groove may be particularly selected based on one or more operating conditions for the fluid end.

(23) FIG. 1 is a schematic cross-sectional view of an embodiment of a pump assembly **100**, which may also be referred to as a reciprocating pump assembly and/or a reciprocating pump. The pump assembly **100** may be utilized during hydraulic fracturing operations, among other operations, where a working fluid (e.g., fracturing fluid, slurry, etc.) is introduced into the pump and energy is added to the working fluid to increase a pressure of the working fluid. Fracturing fluid, by way of example only, may include corrosives and also particulates, such as sand or ceramics, which are utilized during fracturing operations. These corrosives and particulates cause erosion within the pump assembly **100**, which may undesirably affect fracturing operations and lead to down times to replace various components. Additionally, the fracturing fluids may include corrosive acids and the like, which may wear down components of the pump assembly **100**.

(24) It should be appreciated that various components of the pump assembly **100** have been removed for clarity with the following discussion. For example, a power end has been removed in favor of focusing on the illustrated fluid end **102** of the pump assembly **100**. The power end may include a crankshaft that is driven by an engine or motor to facilitate operations. The fluid end **102** includes a fluid end block **104** that may house one or more components discussed herein. A plunger rod **106** is driven (e.g., via the crankshaft) to reciprocate within the fluid end block **104** along a plunger axis **108**. The plunger rod **106** is positioned within a bore **110** extending through at least a portion of the fluid end block **104**. The illustrated bore **110** is arranged along the plunger axis **108** (e.g., first axis) and intersects a pressure chamber **112**, which is arranged along a pressure chamber axis **114** (e.g., second axis), which is positioned substantially perpendicular to the plunger axis **108**. It should be appreciated that the pump assembly **100** may include multiple plunger rod and pressure chamber arrangements, which may be referred to as a plunger throw. For example, the pump assembly **100** may be a triplex pump, quadplex pump, quintuplex pump, and the like.

(25) The illustrated fluid end block **104** includes an inlet passage **116** and an outlet chamber **118**, which are generally coaxial and arranged along the pressure chamber axis **114**. In other words, the inlet passage **116** and the outlet chamber **118** are axially aligned with respect to one another and/or the pressure chamber **112**. In various embodiments, fluid enters the pressure chamber **112** via the inlet passage **116**, for example on an up stroke of the plunger rod **106**, and is driven out of an outlet passage **120**, for example on a down stroke of the plunger **106**.

(26) Respective valve assemblies **122**, **124** are arranged within the inlet passage **116** and the outlet chamber **118**. These valve assemblies **122**, **124** are spring loaded in the illustrated embodiment, but

it should be appreciated that such an arrangement is for illustrative purposes only. In operation, a differential pressure may drive movement of the valve assemblies. For example, as the plunger rod **106** is on the upstroke, pressure at the inlet passage **116** may overcome the spring force of the valve assembly **122**, thereby driving fluid into the pressure chamber **112**. However, on the down stroke, the valve assembly **122** may be driven to a closed position, while the spring force of the valve assembly **124** is overcome, thereby enabling the fluid to exit via the outlet passage **120**.

(27) Piston pumps or plunger pumps, such as the pump assembly **100** shown in FIG. **1**, are positive displacement pumps and are commonly used in environments where the fluids that are being handled pose problems such as high temperatures, viscous media, or solids-charged liquids. One such example is in oil and gas operations, particularly fracturing operations, where solids laden fluids may be used. Examples of these fluids include drilling fluids, muds, cement slurries, fracturing slurries, acids and the like, which frequently must be pumped under high pressure into the well. These abrasive fluids provide challenges for the various sealing interfaces of the pumps. One such interface includes along the plunger **106**. As will be described, various systems and methods of the present disclosure may include a stuffing box configuration that includes one or more rings to secure a packing material along the bore **110** to block leakage of fluid out of the fluid end **102**. In various embodiments, the one or more rings may be arranged external to the bore **110**, such as against a face of the fluid end **102**.

(28) FIG. **2** is a cross-sectional side view of an embodiment of a stuffing box assembly **200** that may be utilized with one or more reciprocating components, such as the plunger **106** illustrated in FIG. **1**. As noted, like numerals may be used throughout to represent like features, but such numbering for convenience and is not intended to limit the scope of the present disclosure. In this example, the stuffing box assembly **200** is positioned on a back side **202** of the fluid end **102**. It should be appreciated that “back side” is a relative term and is not intended to limit one or more orientations of the fluid end **102**. The plunger **106** extends within the bore **110** along the plunger axis **108**, as indicated above, and may be driven to reciprocate along the plunger axis **108** to add energy to the pressure chamber **112** (FIG. **1**).

(29) The illustrated stuffing box assembly **200** is positioned against a recessed area **204** (e.g., grooved area, offset area, depressed area, etc.) formed in the back side **202**. In at least one embodiment, at least a portion of the stuffing box assembly **200** may extend at least partially into a recessed portion or cut out section formed with respect to the recessed area **204**. In other words, the recessed area **204** represents a removal of material from the fluid end **102** and/or a forming process that provides an area with a reduced lateral position (with respect to the orientation of FIG. **2**) compared to an adjacent area. In this example, the recessed area **204** includes a wall **206** forming the recessed area **204**, which may be a continuous, annular wall **206**. It should be appreciated that reference to a single wall is not intended to be limiting, as there may be various regions formed that have different radial extents or the like to prevent rotation and, moreover, the wall **206** may not be continuous in all embodiments. As an example, in one or more embodiments, the wall **206** extends for a length of the fluid end **102**. However, in other examples, the wall **206** may extend for a particular extent. In one or more embodiments, the recessed portion and/or groove **206** may be omitted and the stuffing box assembly **200** may be pressed directly against the back side **202**, for example along a planar region of the back side **202**. By way of example only, a number of different grooves **204** may be formed along the back side **202**, for example at each bore **110**. In this example, the recessed area **204** receives a pressure ring **208** (e.g., a first ring, a pressure-containing ring) that abuts to the recessed area **204** such that at least a portion of the pressure ring **208** is overlapped by at least a portion of the wall **206**. That is, the pressure ring **208** abuts against the recessed area **204** and, in various embodiments, at least a portion of the pressure ring **208** extends axially out of recessed portion associated with the recessed area **204** (e.g., axially along the plunger **108** in a direction away from the back side **202**). It should be appreciated that a distance of axial extension may be particularly selected based on design conditions, and moreover, that a recessed area depth

may also be particularly selected based on design conditions. As noted, in various embodiments, the recessed area **204** may be omitted such that the stuffing box assembly **200** is flush against the back side **202**. The illustrated pressure ring **208** further includes a pressure ring bore **210** that is axially aligned with the bore **110** to enable reciprocation of the plunger **106** along the plunger axis **108**. However, as will be described below, various dimensions of the pressure ring bore **210** may differ from the bore **110** to facilitate installation of various materials, such as one or more packing assemblies or components thereof.

(30) In the example of FIG. 2, an inner diameter **212** of the pressure ring **208** is greater than an inner diameter **214** of the bore **110**, thereby providing a shelf **216** for packing materials **218** (e.g., stuffing material, a packing assembly). The packing materials **218** include various components, such as seals, brass, and the like, which may be utilized to block fluid flow upward and along the plunger rod **106**. The packing materials **218** may be particularly selected based, at least in part, on one or more operating conditions of the fluid end **102**, as well as various dimensions of the shelf **216**, plunger **106**, and pressure ring **208**, among other components. Accordingly, the illustrated configuration is for example purposes only, and it should be appreciated that more or less material may be included in embodiment where additional or less space is provided for the packing materials **218**. It should further be appreciated that, in various embodiments, the shelf **216** may be formed, at least in part, by the pressure ring **208**. For example, the inner diameter **212** may not be constant across the pressure ring **208** and may include the shelf **216** at a variable diameter portion. That is, the pressure ring **208** may include a first inner diameter at the inner diameter **212** and a second inner diameter, less than the first inner diameter **212**, that forms the shelf. However, in various other embodiments, the first diameter **212** (e.g., the portion that engages the packing materials **218**) may be smaller than a second inner diameter. Furthermore, one or more components, such as a spacer ring, may also be incorporated to serve as the shelf **216**.

(31) In various embodiments, an adjacent ring **220** (e.g., outer ring, second ring, non-pressure containing ring, etc.) is positioned to abut the pressure ring **208**. In this example, the adjacent ring **220** is at least partially overlapped by the pressure ring **208**. That is, an overlapped region **222** includes a second pressure ring inner diameter **224**, larger than the inner diameter **212**, that receives a lip **226** of the adjacent ring **220**. As a result, a portion of the pressure ring **208** is illustrated as radially overlapping a portion of the adjacent ring **220**. In various embodiments, the lip **226** is proximate a seal groove **228** formed in the pressure ring **208** that receives a seal **230**. It should be appreciated that the seal groove **228** and the position of the seal **230** may also be within the adjacent ring **220** in other embodiments or may be partially within the pressure ring **208** and partially within the adjacent ring **220**.

(32) In various embodiments, the overlapped region **222** is configured to be axially farther from the back side **202** (e.g., toward the right relative to the orientation of FIG. 2) with respect to one or more packing materials **218** such that the adjacent ring **220** is not exposed to pressurized fluid from the fluid end **102**. It should be appreciated that the overlapped region **222** may also be positioned along any portion of the packing materials **218** and that the configuration in FIG. 2 is for illustrative purposes only.

(33) The adjacent ring **220** further includes a coupling region **232**, which includes threads in this embodiment, to receive a retainer **234** (e.g., packing nut), which may include mating threads or the like. The retainer **234** may be utilized to maintain a position of the packing materials **218**. It should be appreciated that the retainer **234** may be coupled to the adjacent ring **220** and/or to other components in a variety of ways, such as external fittings, clamps, and the like. As shown, the coupling region **232** is arranged at a first end **236** of the adjacent ring **220**, opposite a second end **238**. The first end **236** is farther from the back side **202** than the second end **238**. The first end **236** further includes a stepped portion **240** such that a stepped portion length **242** is less than an adjacent ring length **244**. The stepped portion **240** provides a region to receive one or more fasteners to couple the adjacent ring **220** to the pressure ring **208**. For example, fasteners may

extend through aligned apertures to join the components together. As illustrated, the stepped portion **240** positions the fasteners out of alignment with the retainer **234** to provide improved access to both components. In other words, the fasteners are arranged axially closer to the back side **202** than a top of the retainer **234** to facilitate access to each of the components. It should be appreciated that the fasteners illustrated are for illustrative purposes only and that other coupling devices may be utilized to join the pressure ring **208** to the adjacent ring **220**.

(34) Returning to the pressure ring **208**, a weep hole **246** is illustrated extending radially through the pressure ring **208**. In this example, the weep hole **246** intersects an axial hole **248** extending axially through the pressure ring **208**. It should be appreciated that one or more of the weep hole **246** or the axial hole **248** may or may not extend entirely through the pressure ring **208**. For example, one or more of the holes **246**, **248** may extend to a particular region to form a blind. In this example, the weep hole **246** is positioned proximate the overlapping region **222**, but various other locations may also include weep holes **246**. That is, the weep holes **246** may be arranged at different locations along the axial length of the pressure ring **208**. Furthermore, one or more weep holes **246** may be drilled circumferentially about the pressure ring **208** that intersect with one or more axial holes **248** or that do not intersect axial holes **248**. By way of example, the pressure ring **208** may include multiple weep holes **246** arranged at different circumferential positions of the pressure ring **208** and one or more of these weep holes **246** may or may not intersect one or more additional axial holes **248**.

(35) The illustrated adjacent ring **220** also includes an injection port **250**, which may be utilized to direct grease toward the packing material **218**. It should be appreciated that the injection port may be arranged at any reasonable location to enable injection of various substances toward the bore **110**. In various embodiments, it may be preferential to position the injection port **250** downstream of the seal **230** such that the injection port **250** is not subject to the pressure of the fracturing fluid.

(36) In operation, the pressure ring **208** and the adjacent ring **220** are coupled to the fluid end **102**, for example via fasteners. As illustrated, the back side **202** may include a face seal **252** between the back side **202** and the pressure ring **208**. In this example, a face groove **254** is formed in the back side **202** to receive the face seal **252**. It should be appreciated that, in other embodiments, the face seal **252** may be associated with the pressure ring **208**, rather than the back side **202**. For example, the face groove **254** may be formed in the pressure ring **208**. Additionally, in at least one embodiment, the face groove **254** may be formed partially within the back side **202** and partially within the pressure ring **208**. In one or more embodiments, the face seal **252** may seal against pressurized fluid within the fluid end **102**. As shown, embodiments provide the packing materials **218** of the associated stuffing box assembly **200** abutting on the fluid end **102**, rather than a sleeve being within the bore, as noted above.

(37) Various embodiments of the present disclosure are directed toward a substantially external stuffing box assembly **200** that is not positioned within the bore **110**, however, it should be appreciated that various alternative configurations may include the stuff box assembly **200** or portions thereof that extend into the bore **110**. In this example, the groove **204** is formed within the back side **202** such that the groove **204** extends a groove depth **256** into the back side **202**. The groove depth **256**, as noted above, may be particularly selected based on operating conditions, tooling conditions, or the like. The illustrated groove **204** is formed with a groove height **258** that is configured to receive the pressure ring **208** with a secure fit such that the groove height **258** may be substantially equal to a pressure ring outer diameter **260**. In embodiments, the configuration of the dimensions may facilitate alignment and installation of the components, where a misalignment may be visually established by resistance of fitting the pressure ring **208** into the groove **204**. Other embodiments may include configurations where the groove height **258** is greater than the outer diameter **260**.

(38) As described above, various embodiments position the stuffing box assembly **200** external to the bore **110**. In the illustrated embodiment, the pressure ring **208** extends axially away from the

back side **202** such that at least a portion of a pressure ring thickness **262** is external the groove **204**. That is, the pressure ring thickness **262** may include an internal portion **264** that is within the groove **204** and an external portion **266** that is outside of the groove **204**. In various embodiments, the internal portion **264** is less than the external portion **266**, but it should be appreciated that a variety of configurations may be utilized within the scope of the present disclosure (e.g., internal greater than external, portions equal, etc.). Accordingly, the pressure ring **208** may be fixed to the fluid end **102** via fasteners **268** that extend through the pressure ring **208** and the adjacent ring **220**. In one or more embodiments, the fasteners **268** include threaded fasteners, such as bolts, but it should be appreciated that additional fasteners may also be used within the scope of the present disclosure, such as clamps, radial fasteners (as opposed to the illustrated axial fasteners), integrated bayonet fittings, and the like.

(39) In various embodiments, the back side **202** may also be referred to as having a fluid end face **270**, which may correspond to the surface within the groove **204**. As a result, in the illustrated example, the pressure ring **208** would abut or otherwise be positioned against the fluid end face **270**. Moreover, an outer face **272** may be used to describe the region around the groove **204**, where the fluid end face **270** is axially closer to the internal components of the fluid end **102** than the outer face **272**.

(40) FIG. 3 is an isometric view of an embodiment of the stuffing box assembly **200**. In this example, the pressure ring **208** and the adjacent ring **220** are both illustrated extending axially away from the back side **202**. Further illustrated are the fasteners **268** that extend through the pressure ring **208** and the adjacent ring **220** to couple the stuffing box assembly **200** to the back side **202**. For example, the fasteners **268** may extend into mating apertures formed in the fluid end **102**. Moreover, as noted above, the retainer **234** is arranged within the adjacent ring **220**, for example coupled via threads. The injection port **250** is arranged positioned proximate the weep hole **246**. As noted above, while a single port **250** and single weep hole **246** are illustrated in this example, more may be included, for example at different circumferential positions.

(41) FIG. 4 is a cross-sectional view of an embodiment of a stuffing box assembly **400** that may share one or more features with the stuffing box assembly **200** of FIG. 2. For simplicity and clarity, certain portions will not be repeated herein. In this example, the pressure ring **208** is arranged within the recessed area **204** and abuts the back side **202** of the fluid end **102**. As described above, the internal portion **264** extends into the groove depth **256**. In this example, the internal portion **264** is less than the external portion **266**. In at least one embodiment, the pressure ring **208** compresses the face seal **252**, which in this example is shown associated with the back side **202** within the face groove **254**, but in other examples may be associated with the pressure ring **208**. Furthermore, as described above, the adjacent ring **220** is coupled to the pressure ring **208**.

(42) In this example, a spacer ring **402** is positioned along a transition **404** at the bore **110**. As shown, the spacer ring **402** is arranged proximate the face seal **252** and extends axially away from the back side **202** such that the spacer ring **402** at least partially overlaps the face groove **254**. It should be appreciated that, in various embodiments, a conjoined groove or set of grooves may be utilized to receive the face seal **252** and the spacer ring **402**. Moreover, in one or more embodiments, material may separate grooves for the face seal **242** and the spacer ring **402**. In various embodiments, the spacer ring **402** may even form a portion of the bore **110**. The spacer ring **402** may be utilized as the shelf **216**, as described above, to receive the packing materials **218**.

(43) As shown, the pressure ring **208** may include a notch **406** that receives a portion of the spacer ring **402**. For example, the notch **406** may represent a groove or an increased diameter portion **408**, compared to the inner diameter **212**, to receive and support at least a portion of the spacer ring **402**. In various embodiments, the pressure ring **208**, in combination with one or more other elements, at least partially blocks axial movement of the spacer ring **402** away from the back side **202** (toward the right relative to the orientation of FIG. 4). It should be appreciated that relative dimensions for the spacer ring **402** and/or notch **406** may be particularly selected based on operating conditions. In

operation, the illustrated stuffing box assembly **400** enables isolation of the pressurized fluids within the fluid end **102** by supporting one or more components against the back side **202** of the fluid end **102**.

(44) As noted above, in this example the pressure ring **208** includes three different internal diameter positions, namely the inner diameter **212**, the second inner diameter **224**, and the increased diameter portion **408**. In this example, the increased inner diameter portion **408** is smaller than the second inner diameter **224**, but larger than the inner diameter **212**. Accordingly, it should be appreciated that various different stepped portions may be positioned along the pressure ring bore **210** to facilitate positioning and retention of one or more components utilized with the stuffing box assembly **200**.

(45) FIG. 5 is an isometric view of an embodiment of the stuffing box assembly **400**. In this example, the pressure ring **208** and the adjacent ring **220** are both illustrated extending axially away from the back side **202**. Further illustrated are the fasteners **268** extending through the pressure ring **208** and the adjacent ring **220** to couple the stuffing box assembly **400** to the back side **202**. For example, the fasteners **268** may extend into mating apertures formed in the fluid end **102**. Moreover, as noted above, the retainer **234** is arranged within the adjacent ring **220**, for example coupled via threads. The injection port **250** is arranged positioned proximate the weep hole **246**. As noted above, while a single port **250** and single weep hole **246** are illustrated in this example, more may be included, for example at different circumferential positions.

(46) FIG. 6 is a cross-sectional view of an embodiment of the stuffing box assembly **200** coupled to a back side **202** of the fluid end **102**. In this example, the recessed area **204** is represented by a cut out region **600**, which will be described in more detail below. In this example, the cut out region **600** includes a sloped entry **602** and a mating face **604**, which represents at least a portion of the back side **202** of the fluid end **102**. As shown, the cut out region **600** includes a depth **606** (similar to the groove depth **256**), which may be representative of removed material or the part may be cast or otherwise formed without this material. As a result, inclusion of additional components abutting the back side **202** may not substantially increase an overall size of the fluid end **102**. While the illustrated embodiment includes the recessed area **204** as the cut out region **600**, it should be appreciated that various embodiments may include an extruded or outwardly extended region as opposed to a cut out region. By way of example, the portions illustrated as the cutout region **600** may extend outwardly and away such from back side **202** as opposed as being recessed. Moreover, as will be described below, additional geometries may also be utilized to form the cut out region **600** and/or any extruded or outwardly extending region.

(47) It should be appreciated that the sloped entry **602** is for illustrative purposes only, along with the substantially symmetrical configuration of the cut out region **600**. For example, the sloped entry **602** may be replaced by squared (e.g., approximately 90 degree) walls and/or a curved entry. Furthermore, in various embodiments, the cut out region **600** extends along a length of the fluid end, as illustrated in below. However, in other embodiments, individual cut out regions **600** may be formed from the fluid end. In operation, the stuffing box assembly **200** may couple to the back end **202**, within the cut out region **600**, to compress the face seal **252**, as described above.

(48) FIG. 7 is a schematic view of an embodiment of the fluid end **102** illustrating the cut out region **600**. In this example, the cut out region **600** is substantially squared. That is, the sloped entry **602** (FIG. 6) is replaced by a squared entry **700**, which may be substantially 90 degrees. Like the sloped entry **602**, the squared entry **700** may be referred to as having a polygonal profile. The squared entry **700** transitions to the depth **606**, as noted above. Moreover, the cut out region **600** extends for a fluid end length **702**, but it should be appreciated that the cut out region **600** may be segmented or not extend the full length **702**. The illustrated depth **606** is also particularly selected based on operating conditions.

(49) In one or more embodiments, the fluid end **102** is forged or otherwise manufactured in the configuration shown, which may result in less material being utilized in operations. In one or more

embodiments, the fluid end **102** may also be machined to include the cut out region **600**.

Accordingly, the general shape of the cut out region **600** may also be selected, at least in part, with manufacturing considerations in mind.

(50) FIG. **8** is a schematic view of an embodiment of the fluid end **102** illustrating the cut out region **600**. In this example, the sloped entry **602** is visible, as shown in FIG. **6**. Moreover, the cut out region **600** extends for a fluid end length **702**, but it should be appreciated that the cut out region **600** may be segmented or not extend the full length **702**. The illustrated depth **606** is also particularly selected based on operating conditions. This example further includes the face groove **254** for the face seal **252** (FIG. **2**) and the apertures for receiving the fasteners **268** (FIG. **2**).

(51) FIG. **9** is an isometric view of an embodiment of the fluid end **102** including the stuffing box assemblies **200** coupled to the back side **202** within the cut out region **600**. In this example, the cut out region **600** extends for the fluid end length **702**. As noted above, in various embodiments of the present disclosure, stuffing box assemblies **200** are arranged such that one or more components bear against the back side **202** of the fluid end **102** and without extending into the bore **100**.

(52) It should be appreciated that various embodiments of the present disclosure may incorporate materials formed from similar or dissimilar materials. By way of example, the pressure ring **208** may be formed from a stronger, more resistant material than the adjacent ring **220** due to the pressure ring **208** being in contact with pressurized fluid while the adjacent ring **220** may be isolated from the fluid. Accordingly, less expensive materials may be utilized for the adjacent ring **220**. It should be appreciated that all materials used may be particularly selected for expected operating conditions.

(53) FIG. **10** is a bottom perspective view of an embodiment of the fluid end **102** including the cutout region **600** having a sinusoidal pattern **1000**, which may also be referred to as having a curved or arcuate profile. As noted above, while various embodiments may include the squared entry and/or the sloped entry, a variety of different patterns may be included for forming the cutout region **600**. In this example, the sinusoidal pattern **1000** extends along the length **702** of the fluid end **102**. As noted above, in various embodiments of the present disclosure, stuffing box assemblies **200** are arranged such that one or more components bear against the back side **202** of the fluid end **102** and without extending into the bore **100**.

(54) FIGS. **11** and **12** are cross-sectional views of an embodiment of the fluid end **102** including the stuffing box assembly **200** in which the pressure ring **208** is integrally formed with and/or is a continuous portion of the fluid end block **104**. In this example, the pressure ring **208** extends axially out of the cut out region **600** and may replace or otherwise incorporate the mating face **604** (FIG. **6**) described above with a continuous material layout such that the pressure ring **208** is formed from a common material with respect to the fluid end block **104**.

(55) As noted above, the pressure ring **208** includes the pressure ring bore **210**, which has a larger diameter (e.g., diameter **224**) than the bore diameter **214**. In this manner, the shelf **216** may be formed along the fluid end block **104** to receive and support the packing materials **218**, as noted above. Accordingly, the packing materials **218** may be described as not extending into the bore **110** of the fluid end block **104** and being axially blocked or restricted from moving into the bore **110** by the shelf **216**. Further illustrated in FIGS. **11** and **12** are the seal groove **228** and seal **230** between the pressure ring **208** and the adjacent ring **220**.

(56) As will be appreciated, various features of FIG. **2** are shared in FIGS. **11** and **12**, including the configuration of the adjacent ring **220** with respect to the pressure ring **208**. For example, in this example, a lip **226** of the adjacent ring **220** may extend axially toward the bore **110** such that a portion of the pressure ring **208** radially overlaps at least a portion of the adjacent ring **220**. This may be referred to as the overlapped region **222**, as noted above, where the pressure ring **208** includes the diameter **224** to accommodate the lip **226**. Further shown is the retainer **234** extending into the adjacent ring **220** to secure the packing materials **218** in place.

(57) Various embodiments of the present disclosure illustrate a secure coupling between the

adjacent ring **220** and the pressure ring **208** and/or the fluid end block **104** via one or more fasteners **300**, which may extend through apertures or holes through the adjacent ring **220** and/or pressure ring **208** and/or fluid end block **104** that are aligned to permit passage of the fasteners **300**. As noted above, fasteners **300** are provided by way of example only and various other components may be utilized to join different features together, such as press fittings, interference fits, adhesives, bayonet connections, tongue and groove fittings, and the like.

(58) Embodiments of the present disclosure further illustrate an arrangement that permits easier access to different components of the stuffing box assembly **200** due to the improved clearance between components. For example, as noted above, the pressure ring **208** may extend axially away from the back side **202** such that at least a portion of the pressure ring thickness **262** is external the groove **204**. That is, the pressure ring thickness **262** may include the internal portion **264** that is within the groove **204** and the external portion **266** that is outside of the groove **204**. In various embodiments, the internal portion **264** is less than the external portion **266**, but it should be appreciated that a variety of configurations may be utilized within the scope of the present disclosure (e.g., internal greater than external, portions equal, etc.). When combined with the groove **204**, which may have a depth **606** (FIG. 6), additional clearance area may be provided to various components, such as the fasteners **300**, thereby providing improved access for maintenance or other operations, such as packing replacement and the like.

(59) FIG. 13 is a sectional perspective view of an embodiment of the stuffing box assembly **200** illustrating the pressure ring **208** formed as an integral feature of the fluid end block **104**. In this example, the pressure ring **208** extends away from the back side **202**, which may permit increased working area for operators, as shown with respect to the illustrated fasteners **300**. Further illustrated is the arrangement of the adjacent ring **220** axially outward with respect to the back side **202**. As noted, this configuration may permit the adjacent ring **220** to be formed from materials that are less expensive and/or less likely to be in contact with corrosive or otherwise high pressure fluids, for example, due to the arrangement of the packing materials **218**. It should, however, be appreciated that various embodiments may adjust one or more dimensions such that the adjacent ring **220** extends axially closer to the back end **202** such that the packing materials **218** overlap or otherwise are positioned adjacent to the adjacent ring **220**.

(60) FIG. 14 is a perspective view of an embodiment of the stuffing box assembly **200** that illustrates the adjacent ring **220** extending axially away from the back side **202**. Further illustrated in this configuration is the injection port **250**. As shown, the fasteners **300** are arranged circumferentially to extend through the adjacent ring **220**, which may further go through aligned holes in the pressure ring **208** (FIGS. 11-13), which may or may not extend all the way to the fluid end block **104** (e.g., through the entire pressure ring thickness **262** (FIGS. 11 and 12)).

(61) This application is a continuation of U.S. Non-Provisional application Ser. No. 17/864,873, filed Jul. 14, 2022, titled “DUAL RING STUFFING BOX,” which is a continuation-in-part of U.S. patent application Ser. No. 17/213,854, filed Mar. 26, 2021, titled “DUAL RING STUFFING BOX,” now U.S. Pat. No. 11,391,374, issued Jul. 19, 2022, which claims the benefit of U.S. Provisional Patent Application No. 63/137,501, filed Jan. 14, 2021, titled “DUAL RING STUFFING BOX,” and is a continuation-in-part and claims the benefit of U.S. patent application Ser. No. 29/769,314, filed Feb. 4, 2021, titled “FLUID END FOR A PUMPING SYSTEM,” now U.S. Patent No. D933104, issued Oct. 12, 2021, and is a continuation-in-part and claims the benefit of U.S. patent application Ser. No. 29/769,318, filed on Feb. 4, 2021, titled “FLUID END FOR A PUMPING SYSTEM,” now U.S. Patent No. D933105, issued Oct. 12, 2021, the disclosures of which are incorporated herein by reference in their entireties for all purposes.

(62) The foregoing disclosure and description of the disclosed embodiments is illustrative and explanatory of the embodiments of the disclosure. Various changes in the details of the illustrated embodiments can be made within the scope of the appended claims without departing from the true

spirit of the disclosure. The embodiments of the present disclosure should only be limited by the following claims and their legal equivalents.

Claims

1. A fluid end comprising: a fluid end body having a base; a fluid end bore positioned in the fluid end body and having a fluid end bore diameter; a pressure ring extending from the base and having a pressure ring bore, the pressure ring bore having a pressure ring bore diameter larger than the fluid end bore diameter; an adjacent ring connected to the fluid end, the pressure ring positioned between the adjacent ring and the base, the adjacent ring including an aperture; one or more fasteners extending through the aperture of the adjacent ring to connect the adjacent ring to the fluid end, the one or more fasteners extending through the pressure ring and into the base; and packing material positioned within at least a portion of the pressure ring bore, so that axial movement toward the fluid end is inhibited by the base.
2. The fluid end of claim 1, further comprising: a shelf positioned to support the packing material, the shelf located at a transition between the pressure ring bore diameter and the fluid end bore diameter, and wherein axial movement of the packing material further is blocked by the base.
3. The fluid end of claim 1, wherein the pressure ring is integrally formed with the fluid end, and wherein the shelf includes, at least in part, a spacer ring.
4. The fluid end of claim 1, further comprising: a cut out region defining a recessed area positioned to surround the pressure ring, the cut out region extending axially into the fluid end face, toward the base, for a cut out depth, and wherein axial movement of the packing material further is blocked by the base.
5. The fluid end of claim 4, wherein the pressure ring has a pressure ring thickness, wherein the pressure ring thickness includes an internal portion and an external portion, and wherein the internal portion has a thickness approximately equal to the cut out depth.
6. The fluid end of claim 4, wherein the cut out region extends for a length of the fluid end body.
7. The fluid end of claim 1, further comprising: an overlapping region located between the pressure ring and the adjacent ring, and wherein the pressure ring includes a recess to receive a lip of the adjacent ring.
8. The fluid end of claim 1, further comprising: a seal positioned between the pressure ring and the adjacent ring.
9. A fluid end comprising: a fluid end body having a base; a fluid end bore positioned in the fluid end body and having a fluid end bore diameter; a pressure ring extending from the base and having a pressure ring bore, the pressure ring bore having a pressure ring bore diameter larger than the fluid end bore diameter; an adjacent ring connected to the fluid end, the pressure ring positioned between the adjacent ring and the base; one or more fasteners extending through an aperture of the adjacent ring to connect the adjacent ring to the fluid end; packing material positioned within at least a portion of the pressure ring bore, so that axial movement toward the fluid end is inhibited by the base; and a shelf positioned to support the packing material and located at a transition between the pressure ring bore diameter and the fluid end bore diameter.
10. A fluid end comprising: a fluid end body; a recessed area positioned in the fluid end body; a fluid end bore positioned in the fluid end body; a pressure ring extending from the recessed area, the pressure ring having a pressure ring thickness and a pressure ring bore positioned co-axially with the fluid end bore, the pressure ring bore having a diameter greater than the fluid end bore; an adjacent ring positioned against the pressure ring and connected to the fluid end body; one or more fasteners extending through the pressure ring thickness; packing material positioned within the pressure ring bore, at least a portion of the packing material positioned external to the fluid end bore; and an injection port extending through the adjacent ring, the injection port positioned in a low pressure region.

11. The fluid end of claim 10, further comprising a seal positioned between the pressure ring and the adjacent ring, wherein the pressure ring and the adjacent ring are connected together via the one or more fasteners, and wherein the injection port further is positioned axially away from the seal.
 12. The fluid end of claim 10, wherein the pressure ring is integrally formed with the fluid end.
 13. The fluid end of claim 10, wherein the pressure ring and the adjacent ring are at least one of threaded together or connected together via a bayonet connection.
 14. The fluid end of claim 10, further comprising: a shelf located at a transition between the pressure ring bore and the fluid end bore, and wherein the shelf receives and supports the packing material.
 15. The fluid end of claim 10, wherein the recessed area comprises one or more of a polygonal profile or a curved profile.
 16. A method for fluid end assembly, the method comprising: providing a fluid end including a recessed area extending into a fluid end face and a pressure ring extending axially out of the recessed area; positioning an adjacent ring against the pressure ring, the pressure ring having a selected pressure ring thickness and a pressure ring bore; securing the adjacent ring to the fluid end via one or more fasteners extending through the selected pressure ring thickness; and installing packing material into the pressure ring bore.
 17. The method of claim 16, further comprising: connecting a retainer, thereby to compress the packing material against a shelf located at a transition between a fluid end bore and the pressure ring bore.
 18. The method of claim 16, wherein the pressure ring is integrally formed with the fluid end.
 19. The method of claim 16, further comprising: positioning a seal between the pressure ring and the adjacent ring, and connecting the pressure ring and the adjacent ring together via one or more fasteners.
 20. The method of claim 19, wherein the fluid end further includes an injection port located axially away from the seal.
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